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Global Facilities - Design & Construction Standard - CSA - CUB Building Specification

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[ECN History](#)
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1.0 Purpose

- 1.1 This document provides a Required Standard and Standard Guideline on material and design specification that shall be followed and complied in the Design, Engineering and Construction of CUB Building as part of any green and/or brown field construction projects contract requirement.
- 1.2 The standard is applicable for any green and/or brown field construction projects with regards to provide CUB Building as per required in the contract.
- 1.3 The document is intended to serve as a reference and guideline in addition to any Construction Drawings and Specification provided in the contract requirement including any required by local authority regulation and standard requirement as well as code of practice.
- 1.4 The established Specifications Standard for each locale shall also comply with Micron's in-house:
 - 1.4.1 RBA Code of Conduct requirements for Semiconductor Manufacturing
 - 1.4.2 Environmental, Health & Safety Standards
 - 1.4.3 Operational Requirements /SOP (Standard Operation Procedures)
 - 1.4.4 Security Protocol
- 1.5 The Specifications Standard to be adopted in each local shall not violate any local codes or by-laws and reflects the current semi-conductor industry's best practices.
- 1.6 Where there is the option to adopt either the MasterSpec or in-region specifications standard, the AE (Architectural & Engineering Consultants) shall propose the most suitable solution based on safety, quality, time & cost to Micron's Designated Project Team (DPT) for review and acceptance. Otherwise, the more stringent option shall apply.

2.0 Scope

- 2.1 This document for CUB Building specification covers all necessary related design, engineering, procurement, construction, installation and completion of piling, substructure, superstructure, and roof structure, roof finishes, external wall enclosure, internal wall and partition, finishes, Interior Design (ID) works, workstations & cabinets, fixtures and fitting, miscellaneous works to full compliance of the contract requirement and required for the project. It extended to cover any green and/or brown field construction projects.
- 2.2 This document applies to all Micron Facilities employees, contractors, and vendors at all Micron sites.

3.0 Definitions and Acronyms

- 3.1 Definitions of acronyms used in this document are listed as follows:

Term/Acronym	Definition / Description
AE	External Architectural & Engineering Consultants
DPT	Micron's Designated Project Team
EHS	Micron's Environmental Health & Safety Team
FCT	Facilities Central Team. This is used interchangeably with Global Facilities Engineering Team and associated to O&M BKMs
GFC	Global Facilities & Construction Team. This is used interchangeably with Global Facilities (GFac), Facilities Central Group (FCG), World Wide Operations (WWOps), World Wide Facilities (WW_FACILITIES), Corp Facilities
GFCT	Micron's Global Facilities Construction Team
GFTT	Global Facilities Technology Team. This is used interchangeably with Global Facilities Engineering Team and associated to design standards
Global Facilities Central Team	This is used interchangeably with Innovation & Integration Team
Global Facilities Construction Team	This is used interchangeably with Global Construction Team
Global Facilities Operations Support Team	This is used interchangeably with Global Facilities Operations Team
Global Facilities Planning Team	This is used interchangeably with Global Facilities Line Planning Team
QP/PE	External Qualified Person (professional architect/engineer)
SME	Micron's Subject Matter Expert
TM	Micron's Team Members

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4.0 CUB Building Design Requirements

- 4.1 All CUB Building structural design specifications for any green and/or brown field construction project shall comply to local/international codes & regulations.
 - 4.1.1 Design of Concrete Structure SS EN 1992
 - 4.1.2 Design of Steel Structure SS EN 1993
 - 4.1.3 Geotechnical Design-General Rules SS EN1997-1
 - 4.1.4 Action on Structure SS EN 1991
 - 4.1.5 Singapore National Annexes to EC
- 4.2 Design Stresses.
 - 4.2.1 CONCRETE GRADE
Characteristic Strength of Concrete at 28-days:
Bored Pile $f_{ck} = 32 \text{ N/mm}^2$
Pile Caps $f_{ck} = 32 \text{ N/mm}^2$
Reinforced Concrete Structure $f_{ck} = 32 \text{ N/mm}^2$
 - 4.2.2 STEEL REINFORCEMENT
Characteristic Strength of High Yield Bar, $f_{yk} = 500 \text{ N/mm}^2$
Characteristic Strength of Mild Steel Bar, $f_{yk} = 250 \text{ N/mm}^2$
Characteristic Strength of Steel Fabric, $f_{yk} = 485 \text{ N/mm}^2$
 - 4.2.3 STRUCTURAL STEEL
Steel Grade S275
- 4.3 The structural framing system of CUB Building shall be based on the architectural framing space layout taking into consideration all the room requirements by each Micron discipline engineers.
- 4.4 Depending on the area size and specific room assignments, the loading plan should be first defined as per the requirements before sending to AE to design the building. The design process varies from each local code requirements.
- 4.5 AE design includes the authority submission for piling, substructure, superstructure, steel structure works as per local code requirements. The design process varies from each local code requirements.

5.0 CUB Typical Loadings

Loadings may vary as per local code requirements versus Micron standard loadings and to use whichever is higher value for design. Refer to the Global standard for CUB loading - Global Facilities - Design & Construction - CSA - FAB, CUB, and Admin & Probe Structural Loading Specification.

6.0 CUB Building Foundation Design

The foundation design for CUP Building depends to the total structural loading and the ground characteristic as per soil report. In most cases, Micron CUB proposed foundation system for the building is Bored Piles of grade fck = 32 N/mm². Some may opt to use spun piles or other foundation as per local availability or preferred design by AE. The diameter and total number of the piles depends to the total column load taking per gridlines. In evaluating the working capacity of piles, both the material strength and soil capacity of piles were considered.

7.0 CUB Structural System

The structures are reinforced concrete construction; comprise of pre-cast Beams, Columns and Slabs system and conventional beams/slabs floor systems for lower floors. Steel structures are adopted for upper floor like cooling tower.

- 7.1 Level 1 consists of cast in-situ flat plate slabs.
- 7.2 Level 4 Mezzanine consists of bonked slabs supported by structural steel beams and steel columns.
- 7.3 Level 2 consists of pre-cast single and double tee slabs spanning one direction supported by pre-cast main beams.
- 7.4 Level 2 consists of pre-cast single and double tee slabs spanning one direction supported by pre-cast main beams.
- 7.5 Roof Level consists of pre-cast single and double tee slabs spanning one direction supported by pre-cast main beams to support the roof steel framing cooling tower, equipment plinths, dog houses, gutters.
- 7.6 Upper roof level consists of steel roof supported by steel beams.

Vertical elements consisting of pre-cast columns and cast in-situ core walls transfer all induced load to piled foundation.

The effect of wind load and equilateral horizontal force (EHF) required by the code on the building have been considered.

The rigidity and stability of the building is adequately provided by the reinforce concrete frame works and lift core.

8.0 Load Combinations

Load combination in accordance with local structure design guidelines shall be applied to RC frame structure & steel structure analysis respectively to compute design moment & shear.

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9.0 Document Control

9.1 Approval: GLOBAL_FAC_GFTT_APPR

9.2 Notification:

GLOBAL_FAC_MANAGERS
GLOBAL_FAC_COMMUNICATIONS
GLOBAL_FAC_CONSTRUCTION
GLOBAL_FAC_NOTIFY
FCG_F4_FAC_NOTIFY
FCG_F6_FAC_NOTIFY
FCG_F10_FAC_NOTIFY
FCG_F11_FAC_NOTIFY
FCG_F15_FAC_NOTIFY
FCG_F16_FAC_NOTIFY
FCG_MMP_FAC_NOTIFY
FCG_MMY_FAC_NOTIFY
FCG_MSB_FAC_NOTIFY
FCG_MXA_FAC_NOTIFY
FCG_MTB_FAC_NOTIFY

6.3 Notification:

All Micron Team Members at all Sites

6.4 Retention

Adherence to the requirements specified in

[Global Facilities - Document Control Procedures](#)

6.5 Review

Biennially (two years)

10.0 Revision History

Revision #	Section Changed	Details of Changes	Date
0	NA	New Document	08/15/2022